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Language Model-based Context Augmentation for World Knowledge Bases

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Lack of background knowledge about the everyday world is an obstacle on the way to simulate usual situations and their changes. In this paper we present a simple idea for extending common sense knowledge bases for Japanese language by using a language model. We investigate several semantic categories for which specific knowledge is collected with mask prediction functionality of BERT and the polarity calculation with both next sentence prediction and masking with lexicons. We describe the experimental results and analyze the discrepancies between human evaluators and utilized sentiment analyzer.

1. Introduction

One of the main problems of the current AI is that in spite of revolutionary achievements in machine learning, algorithms still have problems with distinguishing causation and correlation. Although these two are problematic also for human beings in many cases, we have a rich observational experience which machines have not yet obtained. This richness of stimuli we perceive is to some extent graspable by machines but making sense of changing and static parts of environments remains a difficult challenge mostly left to the human side. To reduce the chaotic welter of information, we pay attention to what is important (System 1 role in Kahneman's theory [Kahneman 11]), and then more diligently digest the perceived information upon previously collected experiences (System 2). The problem is that this result of life-long learning is not usually stored and constantly stored by current AI systems, therefore "mind simulations" of the world are difficult to perform. Although we believe it is possible and preferable (also as a partial remedy for explainability problem) to store what was learned in form of natural language, it is costly and time consuming to run life-long learners and combine their knowledge. In this paper we hypothesize that the new context-aware language models utilizing attention could be a temporary solution to this problem and play a role of the collection of experiences for a machine to simulate the world. As we are interested in the value alignment problem, we show our preliminary tests to see if our ethical judgement approach [Rzepka 05] can utilize BERT [Devlin 19] to create and augment possible variations of reality based on a simple input. We also test this language model in regard of consequence analysis and show that combining existing lexicons with mask prediction capability could be also useful in our task.

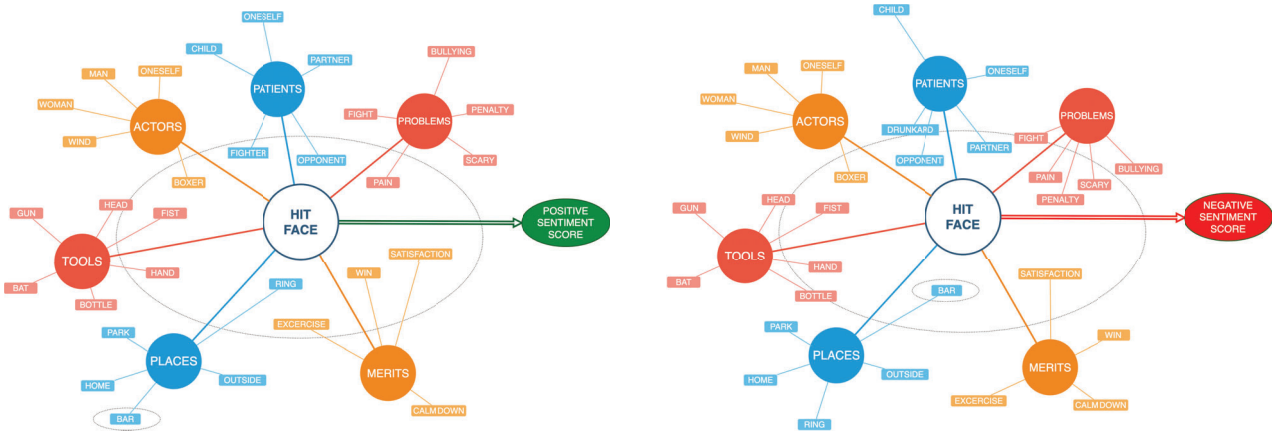
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2. Related Work

Storing world knowledge has a relatively history and the increasing storage capabilities allow us to collect larger and larger datasets. Most of them are for factual data, but there are also several knowledge bases containing commonsense knowledge required for world simulations. One of the iconic projects is Cyc [Lenat 89] aiming at manual or semi-manual creation of world knowledge by trained specialists. MIT's [Singh 02] approach utilized crowdsourcing (students) and later external databases, all cominned later in ConceptNet [Speer 17], which is a graph-based approach, recently followed by ATOMIC [Sap 19a] focusing on inferential knowledge organized as typed if-then relations with variables. The quality of data is ensured by crowd-workers and internal mechanisms. It has been shown in the paper that neural network models that can learn to reason about previously unseen events to generate their likely causes and effects in natural language. Unfortunately ATOMIC, currently the biggest graph of this kind is only for English language and requires vast number of initial concepts to be created, opposite to our proposed method which can create extended knowledge just from single or a pair of input concepts.

Projects such as YAGO [Suchanek 07], KNEXT [Gordon 09], NELL [Carlson 10] or WebChild [Tandon 14] aim at extracting semantic assertions from unstructured text data found on the Internet to avoid human intervention. The problem with evaluating above-mentioned knowledge has led to creating various textual benchmark datasets, most of them for (English) language, some for images [Roemmele 11, Levesque 12, Sap 19b, Zellers 18] which is an important step forward to proper evaluation of the knowledge itself, however such knowledge is very static and discovering subtle contextual changes is still difficult.

Recently automatic augmentation and completion of existing knowledge bases, especially in graphs, have become popular and promising topics [Ma 17, Qian 18, Li 19, Li 19], also using language models like BERT [Yao 19]. Unfortunately works on Japanese language are limited due to the lack of resources.



(a) @Ring (b) @Bar

Figure 1: A concept of "boxer hitting a face" in the working memory assuming a boxing match scenario. Center circle symbolizes current active context and small circle shows a context element to be confirmed as input in the second scenario.

The topic of world simulation is most often related to mirroring physical world in graphical representations, but there are also successful combinations of real-world and text knowledge [Takagi 11, Daruna 19]. As a next step, we plan to implement our ideas to improve the vacuum-cleaning robot's reasoning about the world.

3. Algorithms

The main goal of this preliminary endeavor is to see if a popular language model is capable of augmenting simple concepts, calculating consequences and discovering contextual influencers, i.e. world knowledge elements which can lead to future changes of the situation. We believe this forecasting capability is crucial for an AI to assess its priorities when choosing an action. Figure 1 shows an example of the contextual influencer (*PLACE* in this case) and the sentiment change.

3.1 Knowledge Augmenter

The knowledge augmentation process is the same as in our previous work [Rzepka 20] and the algorithm is shown in Figure 2. As an input, we use pairs of simple events in causal relation retrieved from approximately 100 million Japanese Web pages [Shibata 14]. An input pair from the database is divided into left and right acts A_1 and A_2 stripped to core verbs and then completed by candidates generated by the standard version of Japanese BERT [Shibata 19] (25 most probable candidates) with the masking phrase [MASK]+particle+A. Acquired words were then filtered by JUMAN++[*1] and if the word was a noun with the analyzer's semantic category, it was then searched in YACIS corpus [Ptaszynski 12] for occurrences. If the extended act had no semantic information or was not in the corpus, it was omitted.

3.2 Sentiment Analyzer

In our previous attempts [Rzepka 20], we have tested BERT's next sentence prediction functionality to calculate

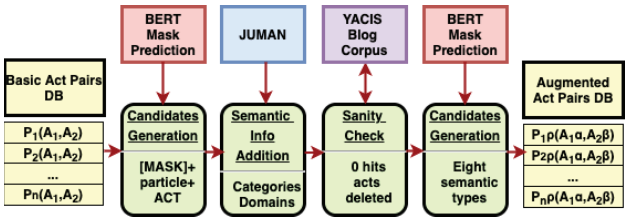


Figure 2: Knowledge augmentation process.

the probability of single positive or negative sentences after the input act. We have tested a set of acts concentrating on a verb *naguru* ("to hit"/"to punch") completed by the algorithm described in 3.1. When compared to an open source sentiment analyzer Asari[*2], the results were similar, but the difference with evaluators score was large. This time we have used a polarity lexicon based on emotional and social consequences [Rzepka 15] to replace the next sentence prediction with mask prediction of BERT. When an act is input, the additional sentence *sore-wa totemo [MASK] desu.* ("This is very [MASK]") and 200 most probable replacement words are then matched with negative and positive words from lexicons. Because probability plays an important role, we have added positional weights to the polarity count and then normalized with minmax scaling to mirror the 0-10 Likert scale used in the human survey presented in [Rzepka 20]. However, this approach did not outperformed previously tested methods with 0.48 error rate vs. next sentence prediction (0.43) and Asari (0.40) which was used fo further comparison. Interestingly, when scores of all three methods were averaged, the standard error dropped to 0.36.

4. Analysis

4.1 Data and Method

To discover what influences the changes in a given context, we compared the differences in semantic categories of

∗1 http://nlp.ist.i.kyoto-u.ac.jp/index.php?JUMAN++

∗2 https://github.com/Hironsan/asari

Table 1: Examples of relatively high agreement between human evaluators (H) and the system (S). The higher score the less ethical the hitting act is evaluated.

Actor	Place	Mean	Patient	Reas.	H	S
child	ring	hand	child		7	7
wrestler	ring	hand	child		10	9
	ring		person		7	6
	street		person		8	6
	dungeon				3	6
	station				8	10
	ring				6	7
child	ring	hand	child		7	7
wrestler	ring	hand	child		10	9
	ring		person		7	6
	street		person		8	6
	hall	spell	character		4	4
	dungeon	spell	character		3	3
	hall	spell	character	attack	3	3
player	hall	spell	character	game	2	3

50 acts used in previous experiments[Rzepka 20]. The sentences vary in length and contextual richness, from simple phrases as "a man hits a child" to "a sumo wrestler hits a child with his hand on the sumo wrestling ring". The authors have not controlled the generated words, only the richness, therefore there are also less natural generations as "to hit a character with a button". We have analyzed which semantic categories (actor, place, mean, patient/object, reason) influenced human scores and analyzed errors of the sentiment analyzer.

4.2 Results

Although the agreement between human evaluators was low (κ=0.072), and sentiment analyzers error rate was high, we decided to see if there are some clear "influencers", categories which can substantially (more than 2 points on the scale) change the sentiment toward an act. Only 50 examples are not sufficient for a solid quantitive experiments, but we have manually confirmed that the order of categories influencing changes are: patient, actor, mean, reason and place.

As an example of the importance of patent category: "hitting with a fist" was given 5 on average, while "hitting a person with a fist" scored 8. On the other hand, the system was mistaken in all examples with 'fist". Examples of more or less correct automatic assessments are shown in Tables 1 and 2.

The data showed human consistency in recognizing that "hitting" act, even if usually an unethical action when another person is the patient, is ethically in opposite to material objects as e.g. a "wall" and all cases were scored as "normal" and the system agreed to that, although estimated that the act is slightly worse if a sumo wrestler is the actor.

Table 2: Examples of low agreement between human evaluators the system.

Actor	Place	Mean	Patient	Reas.	H	S
person			person		8	2
child			wall		7	3
wrestler			wall		7	5
character			wall		0	3
		fist	person		8	2
parent		fist	child	education	9	4
parent	ring	stick	child		9	4
parent	ring	tool	child		9	3

5. Conclusions and Future Work

In our previous work we have shown that the more context is given, the easier it becomes to evaluate common sense data. In this paper we had a closer look to what kind of problems a neural language model can cause when used for augmenting the context and world knowledge for simulations. We have described our approach which utilizes automatically chosen word candidates limited to several semantic categories in order to lead the system to contextually correct knowledge augmentation. However, preliminary tests show that richer contextual data and better sentiment analysis method are required to confirm usability of this approach. As the next step we want to generate larger and more controlled dataset for confirming both quality of augmented concepts and the strengths of contextual influencers.

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References

[Carlson 10] Carlson, A., Betteridge, J., Kisiel, B., Settles, B., Jr., E. R. H., and Mitchell, T. M.: Toward an Architecture for Never-Ending Language Learning, in *Proceedings of the Twenty-Fourth Conference on Artificial Intelligence (AAAI 2010)* (2010)

[Daruna 19] Daruna, A., Liu, W., Kira, Z., and Chetnova, S.: RoboCSE: Robot common sense embedding, in *IEEE International Conference on Robotics and Automation (ICRA 2019)*, pp. 9777–9783 (2019)

[Devlin 19] Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K.: BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding, in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pp. 4171–4186, Minneapolis, Minnesota (2019), Association for Computational Linguistics

[Gordon 09] Gordon, J., Van Durme, B., and Schubert, L.: Weblogs as a Source for Extracting General World Knowledge, in *Proceedings of the Fifth International Conference on Knowledge Capture*, K-CAP '09, pp. 185–186, New York, NY, USA (2009), Association for Computing Machinery

[Kahneman 11] Kahneman, D.: *Thinking, Fast and Slow*, Farrar, Straus and Giroux (2011)

[Lenat 89] Lenat, D. B. and Guha, R. V.: *Building Large Knowledge-Based Systems; Representation and Inference in the Cyc Project*, Addison-Wesley Longman Publishing Co., Inc., Boston, MA, USA, 1st edition (1989)

[Levesque 12] Levesque, H., Davis, E., and Morgenstern, L.: The winograd schema challenge, in *Thirteenth International Conference on the Principles of Knowledge Representation and Reasoning* (2012)

[Li 19] Li, N., Bouraoui, Z., and Schockaert, S.: Ontology completion using graph convolutional networks, in *International Semantic Web Conference*, pp. 435–452Springer (2019)

[Ma 17] Ma, S., Ding, J., Jia, W., Wang, K., and Guo, M.: Transt: Type-based multiple embedding representations for knowledge graph completion, in *Joint European Conference on Machine Learning and Knowledge Discovery in Databases*, pp. 717–733Springer (2017)

[Ptaszynski 12] Ptaszynski, M., Dybala, P., Rzepka, R., Araki, K., and Momouchi, Y.: YACIS: A five-billion-word corpus of Japanese blogs fully annotated with syntactic and affective information, in *Proceedings of The AISB/IACAP World Congress*, pp. 40–49 (2012)

[Qian 18] Qian, W., Fu, C., Zhu, Y., Cai, D., and He, X.: Translating Embeddings for Knowledge Graph Completion with Relation Attention Mechanism., in *IJCAI*, pp. 4286–4292 (2018)

[Roemmele 11] Roemmele, M., Bejan, C. A., and Gordon, A. S.: Choice of plausible alternatives: An evaluation of commonsense causal reasoning, in *2011 AAAI Spring Symposium Series* (2011)

[Rzepka 05] Rzepka, R. and Araki, K.: What Statistics Could Do for Ethics? - The Idea of Common Sense Processing Based Safety Valve, in *Papers from AAAI Fall Symposium on Machine Ethics, FS-05-06*, pp. 85–87 (2005)

[Rzepka 15] Rzepka, R. and Araki, K.: *Rethinking Machine Ethics in the Age of Ubiquitous Technology*, chapter Semantic Analysis of Bloggers Experiences as a Knowledge Source of Average Human Morality, pp. 73–95, Hershey: IGI Global (2015)

[Rzepka 20] Rzepka, R., Takishita, S., Araki, K., et al.: Bacteria Lingualis on BERToids–Concept Expansion for Cognitive Architectures, Technical report, JSAI Special Interest Group for Artificial General Intelligence, SIG-AGI-014-10 (2020)

[Sap 19a] Sap, M., Le Bras, R., Allaway, E., Bhagavatula, C., Lourie, N., Rashkin, H., Roof, B., Smith, N. A., and Choi, Y.: Atomic: An atlas of machine commonsense for if-then reasoning, in *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 33, pp. 3027–3035 (2019)

[Sap 19b] Sap, M., Rashkin, H., Chen, D., LeBras, R., and Choi, Y.: SocialIQA: Commonsense Reasoning about Social Interactions, *arXiv preprint arXiv:1904.09728* (2019)

[Shibata 14] Shibata, T., Kohama, S., and Kurohashi, S.: A Large Scale Database of Strongly-related Events in Japanese, in *LREC* (2014)

[Shibata 19] Shibata, T., Kawahara, D., Hashimoto, C., and Kurohashi, S.: Improving Parsing Accuracy of Japanese Using BERT (in Japanese), in *Proceedings of The 25th Anual Conference of Association for Linguistic Processing (ANLP), pp.205-208* (2019)

[Singh 02] Singh, P., Lin, T., Mueller, E. T., Lim, G., Perkins, T., and Zhu, W. L.: Open Mind Common Sense: Knowledge acquisition from the general public, in *On the move to meaningful internet systems 2002: CoopIS, DOA, and ODBASE*, pp. 1223–1237, Springer (2002)

[Speer 17] Speer, R., Chin, J., and Havasi, C.: ConceptNet 5.5: An Open Multilingual Graph of General Knowledge (2017)

[Suchanek 07] Suchanek, F. M., Kasneci, G., and Weikum, G.: Yago: A Core of Semantic Knowledge Unifying WordNet and Wikipedia, in *Proceedings of the 16th international conference on World Wide Web*, WWW '07, pp. 697–706, New York, NY, USA (2007), ACM

[Takagi 11] Takagi, K., Rzepka, R., and Araki, K.: Just Keep Tweeting, Dear: Web-Mining Methods for Helping a Social Robot Understand User Needs, in *Proceedings of Help Me Help You: Bridging the Gaps in Human-Agent Collaboration*, pp. 60–65, Symposium of AAAI 2011 Spring Symposia (SS-11-05) (2011)

[Tandon 14] Tandon, N., De Melo, G., Suchanek, F., and Weikum, G.: Webchild: Harvesting and organizing commonsense knowledge from the web, in *Proceedings of the 7th ACM international conference on Web search and data mining*, pp. 523–532 (2014)

[Yao 19] Yao, L., Mao, C., and Luo, Y.: KG-BERT: BERT for Knowledge Graph Completion, *arXiv preprint arXiv:1909.03193* (2019)

[Zellers 18] Zellers, R., Bisk, Y., Farhadi, A., and Choi, Y.: From Recognition to Cognition: Visual Commonsense Reasoning (2018)